# SENTHAMIL VALAVAN S

+91 9361577533 | senthamilvalavan456@gmail.com | linkedin.com/in/senthamilvalavan-s | github.com/senthamilvalavan

## PROFESSIONAL SUMMARY

Results-driven Data Scientist specializing in Deep Learning, Computer Vision, and Explainable AI (XAI). Proven expertise in architecting end-to-end ML pipelines, optimizing data workflows, and deploying scalable AI solutions. Adept at leveraging modern AI development environments to accelerate production and deliver actionable business insights for global clients.

## TECHNICAL SKILLS

- Machine Learning & AI: Deep Learning, Explainable AI (Grad-CAM), Computer Vision, Predictive Modeling, NLP.
- Frameworks & Libraries: Python, TensorFlow, Keras, Scikit-learn, Pandas, NumPy.
- Data Analytics & DB: Advanced SQL (CTEs, Window Functions), Data Warehousing, Power BI, ETL Pipelines.
- Modern Dev Tools: Cursor AI, GitHub Copilot, PyCharm, Git/GitHub.

## PROFESSIONAL EXPERIENCE

**Lead Data Scientist | Codeniet | Jan 2026 - Present**

- Spearheaded the AI and data analytics strategy, architecting robust data models and ML pipelines that reduced operational bottlenecks by [X]%.
- Engineered scalable data warehousing frameworks to support business-wide analytics and predictive modeling.
- Collaborated closely with full-stack engineering teams to seamlessly integrate deep learning models into production environments and client-facing web portals.

**Data Scientist & Analyst (Contract/Freelance) | Jan 2025 - Present**

- Delivered custom machine learning models and end-to-end data pipelines for international clients, maintaining a 100% project success rate.
- Optimized complex database operations by writing advanced SQL queries (CTEs, window functions) to transform raw datasets into analysis-ready formats.
- Automated client reporting workflows by developing interactive Power BI dashboards, reducing manual reporting time by over [X]% for stakeholders.

## KEY PROJECTS

**Explainable Deepfake Detection System (XAI) | Academic Capstone**

- Engineered a highly accurate deepfake classification system utilizing a custom 6-module architecture covering automated frame extraction, face detection, and model training.
- Designed and deployed a robust data pipeline to collect, preprocess, and warehouse large-scale video frame datasets for optimized model training.
- Integrated Grad-CAM visualization to unlock Explainable AI (XAI) capabilities, successfully translating complex model decision-making processes to non-technical stakeholders.
- Achieved 93% classification accuracy by leveraging TensorFlow and advanced computer vision techniques.

## EDUCATION & CERTIFICATIONS

- B.Tech in Information Technology | Dr. M.G.R. Educational and Research Institute | 2022-2026 | GPA: 7.18
- Certifications: Python for Data Science (NPTEL, 2024), Data Science Using R (Enlight Wisdom, 2023)